Project Design Phase-I

Solution Architecture

Date	8 March 2026
Team ID	Team169
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	NA

Solution Architecture:

1. Data Collection Layer
Insurance claim data is collected from multiple sources such as customer databases, claim forms, vehicle/property records, medical billing systems, and historical insurance datasets.

2. Data Storage Layer
The collected data is stored in a centralized database or data warehouse where structured and historical claim data can be securely maintained for further analysis.

3. Data Preprocessing Layer
Raw data is cleaned and prepared by handling missing values, removing duplicates, encoding categorical variables, and normalizing numerical features to make the dataset suitable for machine learning models.

4. Feature Engineering Layer
Important features such as claim amount, claim frequency, customer history, accident details, billing patterns, and property value are extracted to improve fraud detection accuracy.

5. Machine Learning Model Layer
Machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, or Gradient Boosting are used to train a fraud detection model that classifies claims as fraudulent or genuine.

6. Fraud Detection Engine
The trained model analyzes incoming insurance claims in real-time and predicts the probability of fraud based on learned patterns.

7. Application / API Layer
A web application or REST API allows insurance companies to submit claims data and receive fraud prediction results from the system.

8. Monitoring and Reporting Layer
Dashboards and reports display suspicious claims, fraud probability scores, and analytics to help investigators and risk managers make data-driven decisions.

9. Security and Access Control
Authentication, authorization, and data encryption mechanisms ensure that sensitive insurance data is protected and accessible only to authorized users.

10. Deployment Layer
The system is deployed on cloud platforms or enterprise servers where it can scale to handle large volumes of insurance claims efficiently.

Example - Solution Architecture Diagram:

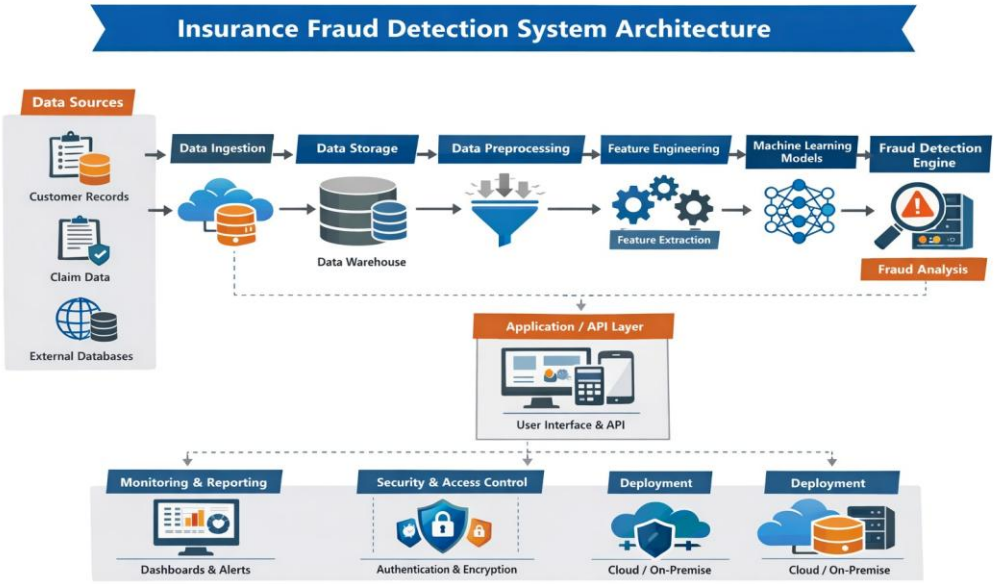

Figure 1: Architecture and data flow of the voice patient diary sample application